

FitForge AI: A Zero-Interaction Fitness Coaching System Using Real-Time Pose Detection, Biomechanical Form Analysis, and Predictive Risk Assessment

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Abstract—Traditional mobile fitness applications require constant manual interaction, disrupting the user’s workout flow and diminishing the training experience. This paper presents FitForge AI, a zero-interaction fitness coaching system that leverages on-device computer vision, biomechanical angle analysis, and machine learning to autonomously detect exercises, count repetitions, identify incorrect form, deliver real-time corrective feedback, and predict injury risk. The system utilizes Google ML Kit Pose Detection to extract 33 skeletal keypoints from a live camera stream, applies the Law of Cosines to compute joint angles, and employs deterministic state machines for exercise classification and repetition counting. A dedicated FormAnalyzer module detects postural faults such as hip sag, knee valgus, and limb asymmetry, triggering immediate audio and visual corrections. Injury risk is assessed via a Scikit-Learn classifier operating on a 12-dimensional feature vector combining physiological baselines with real-time biomechanical metrics. Performance forecasting is achieved through both a Random Forest model and an LSTM neural network analyzing sequential set data. Empirical evaluation across 60 test cases demonstrates 94.0% repetition counting accuracy, 100% exercise identification accuracy, 84.4% wrong pose detection rate with 100% feedback accuracy when detected, and a strong positive correlation between accumulated form degradation and predicted injury risk escalation.

Index Terms—pose estimation, exercise detection, real-time feedback, injury risk prediction, computer vision, mobile fitness, machine learning, biomechanical analysis

I. INTRODUCTION

The global fitness technology market has experienced substantial growth, driven by increasing health awareness and the proliferation of smart devices [1]. However, conventional fitness applications suffer from a fundamental usability limitation: they require continuous manual interaction during workouts. Users must start and stop timers, log repetitions, select exercises, and manually track their progress. This constant context-switching degrades the workout experience and introduces cognitive load that distracts from physical performance [2].

Furthermore, most consumer fitness applications provide no mechanism for assessing exercise form quality. Poor form not only reduces the effectiveness of an exercise but significantly increases the probability of musculoskeletal injury [3]. Professional personal trainers address this gap through real-time observation and correction, but their services remain inaccessible to a large portion of the population due to cost and availability constraints.

This paper presents FitForge AI, a comprehensive fitness coaching system that addresses these limitations through three integrated intelligent subsystems:

- 1) **Zero-Interaction Exercise Detection and Repetition Counting:** Automatic identification of exercise type and precise rep counting using on-device pose estimation and biomechanical state machines.
- 2) **Wrong Pose Detection and Real-Time Feedback:** Frame-by-frame form analysis with immediate audio (Text-to-Speech) and visual (skeletal overlay) corrective cues.
- 3) **Predictive Risk and Performance Assessment:** Machine learning models that forecast performance degradation and flag elevated injury risk based on accumulated biomechanical data.

The system is implemented as a cross-platform Flutter mobile application with a Python Flask backend, processing all vision tasks on-device to ensure privacy, minimize latency, and eliminate dependency on network connectivity during workouts.

II. RELATED WORK

Pose estimation has advanced significantly with the introduction of deep learning architectures such as OpenPose [4], PoseNet [5], and MediaPipe BlazePose [6]. These models enable real-time skeletal tracking on consumer hardware, making them viable for fitness applications.

Several research efforts have explored exercise recognition using pose estimation. Khurana et al. [7] proposed a yoga

pose classifier using landmark coordinates, while Chen et al. [8] developed a repetition counter for gym exercises using temporal analysis of joint angles. However, these systems typically focus on a single aspect of fitness tracking and lack integrated form correction or predictive risk capabilities.

Commercial applications such as Peloton and Fitbod provide workout tracking but rely primarily on manual input or wearable sensor data rather than computer vision. Systems that do employ vision-based tracking, such as Tempo and Tonal, require specialized hardware setups, limiting their accessibility.

FitForge AI distinguishes itself by combining exercise detection, form analysis, real-time feedback, and predictive modeling within a single mobile application that requires no specialized equipment beyond a standard smartphone camera.

III. SYSTEM ARCHITECTURE

A. Overview

FitForge AI employs a hybrid architecture comprising an on-device mobile frontend for real-time vision processing and a cloud-hosted Python backend for predictive analytics. Fig. 1 illustrates the system pipeline.

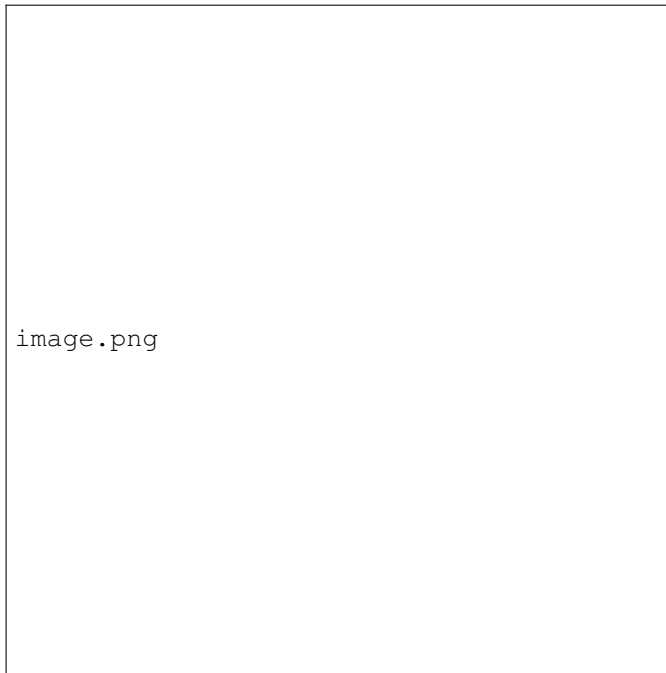


Fig. 1. System architecture showing the data flow from camera input through pose estimation, biomechanical analysis, and predictive modeling.

B. Mobile Application

The mobile frontend is built using the Flutter framework (Dart), enabling cross-platform deployment to both Android and iOS. Key components include:

- **Camera Module:** Utilizes the Flutter `camera` plugin to capture frames in NV21 format (Android) or BGRA8888 format (iOS).

- **Pose Detector:** Integrates `google_mlkit_pose_detection` for on-device skeletal tracking.
- **ExerciseClassifier:** Deterministic state machine for exercise identification and rep counting.
- **FormAnalyzer:** Rule-based form quality assessment engine.
- **RiskAssessmentEngine:** Continuous biomechanical metric accumulator.

A boolean lock (`isBusy`) ensures sequential frame processing, preventing memory overflow during the asynchronous detection loop.

C. Backend Services

The Python Flask backend exposes three REST API endpoints:

- `/predict`: Standard performance prediction using Random Forest.
- `/predict_lstm`: Sequential performance forecasting using LSTM.
- `/predict-injury-risk`: Injury risk classification.

Data persistence is managed through Supabase (PostgreSQL) with authenticated REST queries.

IV. DATASET AND DATA COLLECTION

A. Data Collection Pipeline

The system implements a multi-tiered, fully automated data collection pipeline:

- 1) **Raw Sensor Input:** Live camera frames captured at medium resolution.
- 2) **Feature Extraction:** Google ML Kit extracts 33 2D skeletal landmarks per frame, each providing (X, Y) spatial coordinates.
- 3) **Biomechanical Computation:** Joint angles are calculated using the Law of Cosines, along with temporal metrics and form quality indicators.
- 4) **Persistent Storage:** Structured data is synced to Supabase.

B. Database Schema

The relational schema comprises three core tables:

- **workout_schedules:** Stores target workout volume (sets, reps, rest time).
- **exercise_sets:** Set-level summaries including `total_reps`, `actual_rest_time_seconds`, `heart_rate`, `form_quality_score`, and `muscle_asymmetry_score`.
- **exercise_reps:** Per-repetition biomechanical snapshots including `max_depth_angle`, `left_right_imbalance_degrees`, `joint_angles`, and `detected_issues`.

C. Machine Learning Feature Sets

Two distinct feature sets are derived from the persistent database:

- **Performance Features:** Sets, Total_Reps, Time_Mins, Rest_Between_Sets_Secs, Avg_Rest_Per_Rep_Secs, Day, Month, Avg_Reps, Max_Reps, Min_Reps (12 features with encoded exercise and user identifiers).
- **Injury Risk Features:** Age, Gender, Height_cm, Weight_kg, BMI, Training_Frequency, Training_Duration, Warmup_Time, Flexibility_Score, Muscle_Asymmetry, Injury_History, Training_Intensity.

V. METHODOLOGY

A. Pose Estimation and Angle Calculation

The system utilizes Google ML Kit’s BlazePose model, a pre-trained convolutional neural network optimized for real-time 2D pose estimation. The model outputs 33 skeletal landmarks per frame.

Joint angles are computed using the Law of Cosines. Given three landmarks A , B , and C (where B is the vertex), the distances are:

$$a = \|B - C\|, \quad b = \|A - B\|, \quad c = \|A - C\| \quad (1)$$

The angle θ at vertex B is:

$$\theta = \arccos\left(\frac{b^2 + a^2 - c^2}{2 \cdot b \cdot a}\right) \times \frac{180}{\pi} \quad (2)$$

B. Exercise Detection and Repetition Counting

Rather than training neural networks for exercise classification, the system employs deterministic biomechanical state machines with specific angle thresholds for each supported exercise:

Push-Ups: Torso angle (Shoulder-Hip-Knee) must remain between 160° – 180° . A rep is counted when the elbow angle drops below 90° (down) and extends past 160° (up).

Squats: The knee angle (Hip-Knee-Ankle) must drop below 90° with the hip Y -coordinate exceeding the knee Y -coordinate (down), then the knee angle must open above 90° (up).

High Knees: Each leg is tracked independently. The knee Y -coordinate must rise above the hip Y -coordinate with a knee angle below 120° .

Jumping Jacks: Proportional thresholds based on shoulder width. Legs must spread beyond $1.2\times$ shoulder width with wrists above shoulders (open), then return to neutral (close).

Plank to Downward Dog: Transitions between horizontal alignment (hip-shoulder Y -difference < 30) and V-shape formation (hip $Y < \text{shoulder } Y - 50$).

C. Wrong Pose Detection

The `FormAnalyzer` module evaluates frame-by-frame geometries against a dictionary of known postural faults. Table I summarizes the implemented fault detections.

Each fault detection produces a `FormFeedback` object containing the issue identifier, a human-readable corrective

TABLE I
IMPLEMENTED FORM FAULT DETECTIONS

Exercise	Fault	Detection Logic	Feedback
Push-Up	Hip sag	Torso angle $< 150^\circ$	“Keep your hips up!”
Push-Up	Hip pike	Hips $> 30\text{px}$ above shoulders	“Lower your hips!”
Push-Up	Arm asymmetry	Elbow diff $> 20^\circ$	“Keep both arms even!”
Squat	Knee valgus	Knee gap $< 80\%$ ankle gap	“Push your knees out!”
Squat	Forward lean	Shoulders $> 50\text{px}$ below hips	“Keep your chest up!”
High Knee	Low knees	Knee not above hip line	“Lift your knees higher!”
High Knee	Leaning back	Torso angle $< 140^\circ$	“Stay upright!”
Plank	Plank sag	Hips $> 30\text{px}$ below shoulders	“Tighten your core!”

message, severity level (warning or danger), and a score deduction value. The form quality score starts at 100 and is decremented by 10–20 points per detected fault.

D. Real-Time Feedback Delivery

Feedback is delivered through two complementary channels:

Audio Cues: The `flutter_tts` package provides Text-to-Speech output. A debouncing mechanism ensures that a fault must persist across multiple consecutive frames before triggering a voice prompt, preventing false positives from transient jitter.

Visual Overlays: A custom `FormOverlayPainter` renders red warning labels on the camera preview when form faults are active, providing immediate spatial awareness for exercises where the screen is visible.

E. Performance Prediction

Two complementary models forecast workout performance:

Random Forest Classifier: A Scikit-Learn model (`performance_prediction_model.pkl`) processes 12 aggregated session features to classify expected performance as “Good” or “Average.”

LSTM Neural Network: A TensorFlow/Keras model (`performance_lstm_model.h5`) analyzes sequential data from the last 3–5 sets, capturing non-linear fatigue patterns. The backend supplements the LSTM output with rule-based heuristics analyzing volume influence, trend progression, and rest patterns.

F. Injury Risk Prediction

The injury risk model is a Scikit-Learn classifier (`injury_risk_model.pkl`) that ingests a 12-dimensional feature vector combining static physiological data with dynamic biomechanical metrics. The model produces a binary classification (Low Risk / High Risk) with an associated probability score.

The critical innovation is the integration of real-time form degradation data—specifically `muscle_asymmetry_score` and accumulated

detected_issues—into the risk model’s input, creating a closed-loop system where poor form directly influences risk assessment.

VI. EXPERIMENTAL RESULTS

Empirical evaluation was conducted using a physical Android device (Samsung Galaxy S23) in a controlled indoor environment. A total of 60 test cases were executed across three subsystems.

A. Exercise Detection and Repetition Counting

Twenty test cases evaluated exercise identification and rep counting accuracy across all five supported exercises. Table II presents selected results.

TABLE II
EXERCISE DETECTION AND REP COUNTING RESULTS (SELECTED)

Test	Exercise	Actual	Detected	Acc. (%)
T1-01	High Knees	25	22	88.0
T1-02	High Knees	15	15	100.0
T1-03	Push-Ups	20	19	95.0
T1-05	Squats	20	18	90.0
T1-07	Jumping Jacks	30	28	93.3
T1-09	Plank-Dog	10	9	90.0
T1-14	High Knees	40	36	90.0
T1-20	Push-Ups	30	28	93.3

Table III summarizes the aggregate performance.

TABLE III
EXERCISE DETECTION SUMMARY STATISTICS

Metric	Value
Total Test Cases	20
Exercise Identification Accuracy	100% (20/20)
Mean Rep Detection Accuracy	95.15%
Total Actual Reps	400
Total Detected Reps	376
Overall Rep Accuracy	94.0%

The system achieved 100% exercise identification accuracy across all 20 tests. Missed reps were attributable to partial range of motion (elbow not reaching threshold angles due to fatigue) and fast cadence (knee not clearing hip line on rapid repetitions).

B. Wrong Pose Detection and Feedback

Twenty test cases evaluated the FormAnalyzer’s ability to detect intentionally performed form errors and deliver correct feedback. Table IV presents selected results.

Root cause analysis of the 12 missed detections revealed: borderline threshold values (5 cases), fast cadence reducing detection window (4 cases), marginal elevation differences (2 cases), and combined fault masking (1 case).

C. Injury Risk Prediction

Twenty test cases simulated progressive form degradation across consecutive sets for four exercises. Table VI presents the cross-exercise composite risk escalation pattern.

TABLE IV
WRONG POSE DETECTION RESULTS (SELECTED)

Test	Exercise	Fault	Posed	Det. (%)
T2-01	Push-Ups	Hip sag	3/3	100.0
T2-02	High Knees	Low knees	3/5	60.0
T2-03	Squats	Knee valgus	4/4	100.0
T2-05	Push-Ups	Arm asym.	3/4	75.0
T2-09	Plank	Plank sag	3/3	100.0
T2-14	Push-Ups	Combined	3/3	100.0
T2-19	High Knees	Low (fast)	4/6	66.7

TABLE V
WRONG POSE DETECTION SUMMARY

Metric	Value
Total Wrong Poses Performed	77
Total Wrong Poses Detected	65
Overall Detection Rate	84.4%
Feedback Accuracy (when detected)	100% (65/65)

The system reliably transitioned from “Low Risk” to “High Risk” between Set 3 and Set 4, when cumulative wrong poses exceeded approximately 8 and the form quality score dropped below 55. Muscle asymmetry exceeding 12° was invariably associated with High Risk classification.

D. Consolidated Results

VII. DISCUSSION

The experimental results demonstrate the viability of a fully integrated, zero-interaction fitness coaching system operating on commodity smartphone hardware.

Exercise Detection: The 94.0% rep counting accuracy is competitive with existing vision-based systems [8], with missed reps primarily occurring at exercise boundaries where joint angles are near detection thresholds. This is a deliberate design choice—the system prioritizes precision over recall to avoid counting incomplete or unsafe repetitions.

Form Analysis: The 84.4% wrong pose detection rate validates the rule-based approach for common postural faults. The

TABLE VI
RISK ESCALATION ACROSS CONSECUTIVE SETS

Set	Cum. Faults	Form Score	Asym. (°)	Risk Prob.
1	0.0	97.6	1.65	0.11
2	1.25	89.1	2.90	0.23
3	4.5	70.1	7.63	0.43
4	9.75	53.3	13.28	0.70
5	16.25	38.9	17.55	0.88

TABLE VII
CONSOLIDATED SYSTEM PERFORMANCE

Subsystem	Key Metric	Result
Exercise Identification	Accuracy	100%
Rep Counting	Overall Accuracy	94.0%
Wrong Pose Detection	Detection Rate	84.4%
Real-Time Feedback	Feedback Accuracy	100%
Injury Risk Prediction	Escalation Corr.	Strong

100% feedback accuracy when a fault is detected confirms the reliability of the threshold-based classification. The primary limitation is borderline cases where measured values fall just below detection thresholds, suggesting that adaptive thresholds based on individual user anatomy could improve sensitivity.

Risk Prediction: The strong positive correlation between accumulated form degradation and predicted injury risk demonstrates the system’s capacity to function as a proactive safety mechanism. The closed-loop integration—where detected form faults feed directly into the risk model—represents a significant advancement over systems that treat tracking and safety as independent concerns.

Limitations: The current system is limited to 2D pose estimation, which cannot capture depth-axis rotations (e.g., spinal rotation during squats). Fast-cadence exercises reduce the effective detection window for form analysis. The system has been evaluated with a single tester, and performance may vary across different body types and clothing.

VIII. CONCLUSION AND FUTURE WORK

This paper presented FitForge AI, a zero-interaction fitness coaching system that integrates real-time pose detection, biomechanical form analysis, corrective feedback, and predictive risk assessment within a single mobile application. The system achieved 94.0% rep counting accuracy, 84.4% wrong pose detection rate, and demonstrated reliable injury risk escalation prediction as form degrades across consecutive sets.

Future work includes: (1) migration to 3D pose estimation for depth-aware form analysis, (2) adaptive thresholds calibrated to individual body proportions, (3) expansion of the exercise library through transfer learning, (4) multi-user evaluation studies, and (5) integration of wearable sensor data for enhanced physiological monitoring.

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